Group Segregation Strategies for FTRW:

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Abstract

FTRW (Fairness-Targeted Random Walk) places a limited budget of truth seeds in a social network that is simultaneously exposed to misinformation, with the goal of maximizing the minimum save-probability (maximin welfare) over all vulnerable nodes. In the current implementation, the protected group P is the set of nodes whose save-probability falls below a fixed quantile or threshold. This document proposes four alternative strategies for forming P and the unprotected group U: a statistically-tied minimum band, a rescuability-filtered set, a community-stratified set, and a multiplicative-weights dual persistent group. For each strategy we give the mathematical rep., a SWOT analysis, the rationale for improving the maximin objective, and pseudocode. All strategies use only existing data structures (`sp`, `dist`, `A`) and plug into the Monte-Carlo-driven selection pipeline of `algos/ftrw.py`.

1 Problem Context

The framework in `algos/ftrw.py` follows the methodology in `BTP-ProposedRandomWalkApproach.pdf`. The goal is to allocate a budget k of truth seeds T so as to maximize the minimum save-probability across all *vulnerable* nodes, i.e. nodes that are neither seeded with truth nor with misinformation and whose current save-probability is below 1:

$$\max_{T \subseteq V,\ |T| \le k} \ \min_{v \in V_{\text{vul}}} \ \text{sp}_v(T). \tag{1}$$

1.1 Current FTRW Loop

Repeated until the seed budget is exhausted:

1. **Monte-Carlo estimation** of the save-probability vector $\text{sp}(v)$ for every node, given the current truth seed set T.
2. **Dynamic group formation.** The protected group P consists of the worst-protected vulnerable nodes, i.e. those with $\text{sp}(v)$ below the p_{frac}-quantile (or a fixed threshold θ). The unprotected group U is the remaining vulnerable nodes.
3. **Deficit weights**

$$h(v) = d(v)\left[\eta + (1-\eta)\, e^{-\text{dist}(M,v)/\tau}\right], \qquad d(v) = \text{cut} + \tfrac{1}{R} - \text{sp}(v), \tag{2}$$

   where $\text{dist}(M, v)$ is the shortest-path distance from the misinformation sources M, and τ, η are tunable parameters.
4. **Jump vector v** blending an FSPR-style jump (a fraction φ_{jump} of the restart mass distributed proportionally to $h(v)$ over P) with uniform mass over U and a small exploration term ε.

5. **LFPR-style walk** (or plain Personalized PageRank if $\varphi_{\text{walk}} = \texttt{None}$) on the reversed graph, using $\mathbf{v}$ and the protected-group mask g_P, yielding a score vector π.
6. **Short-listing** of the top-L candidates by π, excluding seeded nodes.
7. **Direct Monte-Carlo evaluation** of each candidate with a common set of random numbers. The candidate with the largest maximin gain (ties broken by the largest gain in mean save-probability of the worst-off nodes) is added to T.

1.2 Auxiliary Components

- Proximity weighting $\text{prox}(v) = e^{-\text{dist}(v)/\tau}$ (already computed).
- Optional degree-based metrics via the sparse adjacency matrix A.
- Directed and undirected graphs are supported; the walk runs on G^{rev} when the graph is directed.
- Hyper-parameters: p_{frac}, θ, φ_{jump}, φ_{walk}, η, τ, ε, γ, `tail_frac`, R_{base}, R_{cand}, etc.

2 Notation

V_{vul}	Set of vulnerable nodes.
$\text{sp}(v)$	Monte-Carlo save-probability estimate of node v using R samples.
s^*	$\min_{v \in V_{\text{vul}}} \text{sp}(v)$, the current minimum; $v^* = \arg\min_v \text{sp}(v)$.
$\text{dist}_M(v)$	Distance from node v to the misinformation set M.
$\text{se}(v)$	Sampling standard error, $\sqrt{\big(\text{sp}(v)(1-\text{sp}(v)) + 1/R\big)/R}$.
$\text{prox}(v)$	Proximity weight $e^{-\text{dist}_M(v)/\tau}$.
$d(v), h(v)$	Deficit and deficit-weight, see Eq. (2).
g_P	Boolean mask of the protected group P.

Every strategy below returns a boolean mask g_P (plus optional soft weights), so it replaces only the quantile block in step 2 without changing steps 3–7.

3 Strategy 1: Statistically-Tied Minimum Band

Idea. Replace the fixed p_{frac}-quantile by "all nodes statistically indistinguishable from the current worst node" (adaptive and noise-aware).

3.1 Mathematicallly can be expressd as

$$P_t = \Big\{ v \in V_{\text{vul}} \;:\; \text{sp}(v) - z\,\text{se}(v) \;\le\; s^* + z\,\text{se}(v^*) \Big\}, \qquad k_{\min} \le |P_t| \le p_{\max}\,|V_{\text{vul}}|. \tag{3}$$

The size constraint is enforced by clipping by sp-rank. *Soft version:* each node gets a weight

$$w(v) = \exp\!\Big(-\tfrac{\text{sp}(v) - s^*}{\beta}\Big), \qquad \beta = \text{median}_v\,\text{se}(v), \tag{4}$$

and the jump mass over P is proportional to $d(v)\,w(v)\,[\eta + (1-\eta)\text{prox}(v)]$.

```
se = np.sqrt((sp*(1-sp) + 1/R)/R)
tie = (sp - z*se) <= (sp[vstar] + z*se[vstar])
gP = tie & vul
# clip to [k_min, p_max*|vul|] by sp-rank
```

3.2 SWOT Analysis

S P shrinks automatically as the minimum is lifted and the sp distribution flattens. It removes the arbitrary p_{frac} and includes only nodes that could plausibly be the true minimum.

W Needs the se estimate and z. The band can be very large early (everyone ties near 0) or tiny late.

O Pairs naturally with $R_{\text{base}}/R_{\text{cand}}$ since larger R tightens the band. The soft version gives a smooth transition between iterations.

T If the sp distribution is heavy-tailed, a few nodes dominate and the walk concentrates on one region. The clip bounds $k_{\min}, p_{\max}$ must be tuned.

3.3 Rationale

Maximin welfare depends only on the true minimum, but with finite R the empirical minimum is a biased-low "winner's curse" estimate. Targeting the tied set aims the walk at every node that could be the bottleneck, rather than chasing noise in the single lowest estimate.

Algorithm 1 FormGroups-TiedBand

Input: sp, V_{vul}, R, z, $k_{\min}$, $p_{\max}$, *soft* flag
Output: Mask g_P (and weights w if *soft*)

$\text{se}(v) \leftarrow \sqrt{(\text{sp}(v)(1-\text{sp}(v)) + 1/R)/R}$ for all $v \in V_{\text{vul}}$
$v^* \leftarrow \arg\min_{v \in V_{\text{vul}}} \text{sp}(v)$; $\quad s^* \leftarrow \text{sp}(v^*)$
$g_P(v) \leftarrow \left[\text{sp}(v) - z\,\text{se}(v) \le s^* + z\,\text{se}(v^*)\right] \wedge [v \in V_{\text{vul}}]$
$n_P \leftarrow |\{v : g_P(v)\}|$
if $n_P < k_{\min}$ **then**
    set $g_P \leftarrow$ the $k_{\min}$ vulnerable nodes with smallest sp
else if $n_P > p_{\max}|V_{\text{vul}}|$ **then**
    set $g_P \leftarrow$ the $\lfloor p_{\max}|V_{\text{vul}}| \rfloor$ vulnerable nodes with smallest sp
end if
if *soft* **then**
    $\beta \leftarrow \text{median}_{v \in V_{\text{vul}}} \text{se}(v)$
    $w(v) \leftarrow \exp(-(\text{sp}(v) - s^*)/\beta)$ for $v \in P$
else
    $w(v) \leftarrow 1$ for $v \in P$
end if
$U \leftarrow V_{\text{vul}} \setminus P$
return $g_P,\ w$

4 Strategy 2: Rescuability-Filtered Protected Set

Idea. Exclude nodes whose sp is structurally pinned (e.g. nodes adjacent to M that misinformation reaches first). They freeze the minimum, so every candidate's maximin gain is about 0 and selection falls back to the tie-break rule.

4.1 Mathematicallly can be expressd as

Define the rescuable set and the protected set:

$$\mathcal{R}_t = \big\{v \in V_{\text{vul}} : \text{dist}_M(v) \geq d_0\big\} \;\cap\; \big\{v : \text{sp}_t(v) - \text{sp}_{t-w}(v) \geq \epsilon_g \;\text{ or } v \text{ is new to } P\big\}, \tag{5}$$

$$P_t = \big\{v \in \mathcal{R}_t : \text{sp}(v) \leq q_p(\text{sp} \,|\, \mathcal{R}_t)\big\}, \qquad U_t = V_{\text{vul}} \setminus P_t. \tag{6}$$

The first condition is a *structural* filter, with $d_0 = 2$ under a synchronous IC-style race (a truth seed at a neighbour can then at best tie the race). The second is a *stagnation* filter using the history of sp across iterations (the sum of per-step increments over a window of length w telescopes to $\text{sp}_t(v) - \text{sp}_{t-w}(v)$). Pinned nodes form a third class $H = V_{\text{vul}} \setminus \mathcal{R}_t$ that is dropped from the jump vector but remains allowed as Monte-Carlo candidates, since seeding them directly is always possible.

4.2 SWOT Analysis

S Prevents budget stalls and uses information already available (`dist`, past `sp`). It is a cheap filter on top of the current pipeline.

W d_0 depends on the diffusion model and tie-breaking rule, and the stagnation window w is another hyper-parameter.

O Makes the selection step meaningful: maximin gain becomes informative again, and the walk focuses on improvable nodes.

T Strictly speaking it relaxes maximin toward leximin, since pinned nodes are deliberately deprioritised. If one pinned node is fixable by a single seed and the budget is large, ignoring it loses the true maximin optimum. *Mitigation:* always include H-nodes in the candidate pool.

4.3 Rationale

When the global minimum cannot move under any one-seed change, optimizing it wastes budget. Filtering for improvability keeps the greedy loop from stalling and lifts the next-worst nodes, which raises the final minimum once the pinned ones are covered.

Algorithm 2 FormGroups-Rescuable

Input: sp_t, history $\{\mathrm{sp}_{t-1}, \ldots, \mathrm{sp}_{t-w}\}$, dist_M, V_{vul}, d_0, w, ϵ_g, p, previous protected set P_{t-1}
Output: g_P, pinned set H

$\mathcal{R} \leftarrow \emptyset$
for each $v \in V_{\mathrm{vul}}$ **do**
  $far \leftarrow [\mathrm{dist}_M(v) \geq d_0]$
  $moving \leftarrow [\mathrm{sp}_t(v) - \mathrm{sp}_{t-w}(v) \geq \epsilon_g] \ \vee \ [v \notin P_{t-1}]$
  if $far \wedge moving$ **then**
    $\mathcal{R} \leftarrow \mathcal{R} \cup \{v\}$
  end if
end for
$H \leftarrow V_{\mathrm{vul}} \setminus \mathcal{R}$
if $\mathcal{R} = \emptyset$ **then** ▷ fallback: no rescuable node
  return FormGroups-QuantileBaseline$(\mathrm{sp}_t, V_{\mathrm{vul}}, p)$, H
end if
$q \leftarrow$ the p-quantile of $\{\mathrm{sp}_t(v) : v \in \mathcal{R}\}$
$g_P(v) \leftarrow [v \in \mathcal{R}] \wedge [\mathrm{sp}_t(v) \leq q]$
$U \leftarrow V_{\mathrm{vul}} \setminus P$
add H to the Monte-Carlo candidate pool (but not to the jump vector)
return g_P, H

5 Strategy 3: Community-Stratified Protected Set

Idea. Select the worst-off nodes *within each community* instead of globally, so that no single region absorbs all the restart mass.

5.1 Mathematicallly can be expressd as

Compute a partition $\{C_1, \ldots, C_K\}$ once from A (Louvain or label propagation). Then

$$P_t = \bigcup_{c=1}^{K} \Big\{ v \in C_c \cap V_{\mathrm{vul}} : \mathrm{sp}(v) \leq q_{p_c}(\mathrm{sp} \,|\, C_c) \Big\}, \qquad p_c = p_{\mathrm{frac}} \cdot \Big(\tfrac{|C_c \cap V_{\mathrm{vul}}|}{|V_{\mathrm{vul}}|} \Big)^{\alpha}. \tag{7}$$

Optionally combine with the global band of Strategy 1: $P = P_{\mathrm{global}}(p/2) \cup P_{\mathrm{strat}}(p/2)$. Rescale the deficit weights per community as

$$\tilde{h}(v) = h(v) \, \frac{\omega_c}{\sum_{u \in P \cap C_c} h(u)}, \qquad \omega_c \propto \Big(\max_{u \in C_c} d(u) \Big)^{\kappa}. \tag{8}$$

5.2 SWOT Analysis

S Cheap (one-time partition) and exploits graph structure in A. It keeps secondary bottlenecks "warm", so the next-worst region is already in P after the worst is lifted.

W Can dilute the target by pulling in healthy nodes from well-protected communities. The result depends on the community detection method and resolution.

O Combines well with the LFPR-style walk, since community-balanced restart mass avoids rank sinks. Provides per-community diagnostics for the report.

T On graphs without clear community structure (near-random or very dense) the partition is arbitrary and the strategy degenerates into a noisier version of the quantile rule.

5.3 Rationale

Personalized-PageRank-style walks concentrate mass near the heaviest part of the jump vector. If the worst nodes sit in one cluster, the walk never scores candidates for the second-worst cluster, and the greedy step lifts only one region at a time, producing a stair-step maximin trajectory. Stratifying evens out the exploration.

Algorithm 3 FormGroups-CommunityStratified

Input: sp, V_{vul}, adjacency A, p_{frac}, α, κ, h, d
Output: g_P, rescaled weights $\tilde{h}$

▷ *One-time preprocessing (cache across iterations)*
if partition $\{C_c\}$ not yet computed **then**
  $\{C_1, \dots, C_K\} \leftarrow$ Louvain(A) (or label propagation)
end if
$g_P \leftarrow \mathbf{0}$
for $c = 1, \dots, K$ **do**
  $V_c \leftarrow C_c \cap V_{\text{vul}}$
  if $V_c = \emptyset$ **then continue**
  end if
  $p_c \leftarrow p_{\text{frac}} \cdot (|V_c|/|V_{\text{vul}}|)^{\alpha}$
  $q_c \leftarrow$ the p_c-quantile of $\{\text{sp}(v) : v \in V_c\}$
  $g_P(v) \leftarrow 1$ for all $v \in V_c$ with $\text{sp}(v) \le q_c$ ▷ at least one node per community
end for
for $c = 1, \dots, K$ **do**
  $\omega_c \leftarrow (\max_{u \in C_c \cap V_{\text{vul}}} d(u))^{\kappa}$; $Z_c \leftarrow \sum_{u \in P \cap C_c} h(u)$
end for
normalise ω so that $\sum_c \omega_c = 1$
$\tilde{h}(v) \leftarrow h(v)\, \omega_{c(v)}/Z_{c(v)}$ for $v \in P$
$U \leftarrow V_{\text{vul}} \setminus P$
return g_P, $\tilde{h}$

6 Strategy 4: Dual-Weight (Multiplicative-Weights) Persistent Groups

Idea. Treat group membership as the *dual variable* of the maximin problem and update it across iterations instead of recomputing it from scratch.

6.1 Mathematicallly can be expressd as

The maximin problem has a Lagrangian form

$$\max_{|T| \le k} \min_{v} \text{sp}_v(T) \;=\; \max_{T} \min_{\lambda \in \Delta(V_{\text{vul}})} \sum_{v} \lambda_v \, \text{sp}_v(T). \tag{9}$$

With entropy regularization the inner minimizer is $\lambda_v \propto \exp(-\text{sp}_v/\beta)$. Maintain λ across iterations with a forgetting factor $\rho \in (0, 1]$:

$$\lambda_{t+1}(v) \;\propto\; \lambda_t(v)^{\rho} \exp\big(-\eta_{\text{mw}}(\text{sp}_t(v) - s_t^*)\big), \quad \text{renormalized over } V_{\text{vul}}. \tag{10}$$

The protected set is the smallest set of nodes (sorted by λ descending) carrying $1 - \alpha$ of the dual mass,

$$P_t = \arg\min\Big\{|S| : \sum_{v \in S} \lambda(v) \ge 1 - \alpha\Big\}, \tag{11}$$

and the jump mass over P is proportional to $\lambda(v)\, d(v)\, [\eta + (1-\eta)\mathrm{prox}(v)]$.

```
lam = lam**rho * np.exp(-eta_mw*(sp - sp[vul].min()))
lam[~vul] = 0; lam /= lam.sum()
order = np.argsort(-lam); cum = np.cumsum(lam[order])
gP = np.zeros(n, bool); gP[order[:np.searchsorted(cum, 1-alpha)+1]] = True
```

6.2 SWOT Analysis

S Principled link to the maximin objective. The memory smooths out Monte-Carlo noise, and nodes that were briefly lifted but are still borderline stay in P.

W Adds state λ and three hyper-parameters ($\eta_{\mathrm{mw}}, \rho, \alpha$); needs careful handling when nodes become seeded or invulnerable.

O Generalizes Strategy 1: the soft band is the one-shot case ($\rho = 0$), which makes ablation easy. It also makes $h(v)$ fully consistent with the dual.

T A high η_{mw} makes λ collapse onto a few nodes (the same concentration failure as a global quantile), and ρ close to 1 makes it slow to react after a seed fundamentally changes the sp landscape.

6.3 Rationale

Greedy on the instantaneous minimum is myopic and noisy. The multiplicative-weights-update (MWU) dual is a standard way to make maximin tractable, and persistent weights stabilize the choice of "who matters". Seeds are then placed for the whole near-bottleneck population rather than whichever node happened to dip lowest in one Monte-Carlo run.

Algorithm 4 FormGroups-MWUDual

Input: sp_t, V_{vul}, persistent λ (init. uniform over V_{vul}), η_{mw}, ρ, α
Output: g_P, updated λ

$s^* \leftarrow \min_{v \in V_{\mathrm{vul}}} \mathrm{sp}_t(v)$
for each $v \in V_{\mathrm{vul}}$ **do**
  $\lambda(v) \leftarrow \lambda(v)^{\rho} \cdot \exp\big(-\eta_{\mathrm{mw}}(\mathrm{sp}_t(v) - s^*)\big)$
end for
$\lambda(v) \leftarrow 0$ for all $v \notin V_{\mathrm{vul}}$ ▷ seeded or invulnerable nodes leave the dual
$\lambda \leftarrow \lambda / \sum_v \lambda(v)$
sort nodes by λ in descending order: $\sigma_1, \sigma_2, \ldots$
$m \leftarrow \min\{\, j : \sum_{i \le j} \lambda(\sigma_i) \ge 1 - \alpha \,\}$
$g_P \leftarrow \mathbf{0}$; $g_P(\sigma_i) \leftarrow 1$ for $i = 1, \ldots, m$
$U \leftarrow V_{\mathrm{vul}} \setminus P$
return g_P, λ

7 Integration into FTRW: Pseudocode

Algorithm 6 shows the full FTRW loop with a pluggable group-formation block. The baseline quantile rule (Algorithm 5) and Algorithms 1–4 all share the same interface and may be swapped via a `group_mode` argument. The jump-vector expression below is one consistent reading of step 4 of the current method (an FSPR-style term over P, a uniform term over U, and an exploration floor ε).

Algorithm 5 FormGroups-QuantileBaseline (current implementation)

Input: sp, V_{vul}, p_{frac} (or threshold θ)
Output: g_P

$q \leftarrow$ the p_{frac}-quantile of $\{\text{sp}(v) : v \in V_{\text{vul}}\}$ (or $q \leftarrow \theta$)
$g_P(v) \leftarrow [v \in V_{\text{vul}}] \wedge [\text{sp}(v) \le q]$
return g_P

Algorithm 6 FTRW with pluggable group formation

Input: Graph $G = (V, E)$, adjacency A, misinformation set M, budget k, group mode $\mathcal{G} \in \{\text{Base}, 1, 2, 3, 4\}$, shortlist size L, hyper-parameters $\eta, \tau, \varepsilon, \varphi_{\text{jump}}, \varphi_{\text{walk}}, \gamma, R_{\text{base}}, R_{\text{cand}}$
Output: Truth seed set T

$T \leftarrow \emptyset$; $\text{dist}_M \leftarrow$ MultiSourceBFS(G, M); $\text{prox}(v) \leftarrow e^{-\text{dist}_M(v)/\tau}$
$G_w \leftarrow G$ if undirected, else G^{rev}
initialise group-formation state (history buffer, partition, λ) as required by $\mathcal{G}$
for $t = 1, \ldots, k$ **do**
  ▷ *1. Estimate save-probabilities*
  $\text{sp} \leftarrow$ MonteCarloSP$(G, M, T, R_{\text{base}})$
  $V_{\text{vul}} \leftarrow \{v \notin T \cup M : \text{sp}(v) < 1\}$
  ▷ *2. Group formation (pluggable)*
  $g_P \leftarrow$ FormGroups$_{\mathcal{G}}(\text{sp}, V_{\text{vul}}, \text{dist}_M, A, \ldots)$; $P \leftarrow \{g_P = 1\}$, $U \leftarrow V_{\text{vul}} \setminus P$
  ▷ *3. Deficit weights*
  $d(v) \leftarrow \text{cut} + 1/R_{\text{base}} - \text{sp}(v)$; $h(v) \leftarrow d(v)\,[\eta + (1-\eta)\text{prox}(v)]$ for $v \in V_{\text{vul}}$
  ▷ *4. Jump vector*
  $\mathbf{v}(u) \leftarrow (1-\varepsilon)\Big[\varphi_{\text{jump}} \dfrac{h(u)\mathbb{1}[u \in P]}{\sum_{x \in P} h(x)} + (1-\varphi_{\text{jump}})\dfrac{\mathbb{1}[u \in U]}{|U|}\Big] + \dfrac{\varepsilon}{|V_{\text{vul}}|}\mathbb{1}[u \in V_{\text{vul}}]$
  ▷ *5. Walk on the (reversed) graph*
  if $\varphi_{\text{walk}} =$ `None` **then**
    $\pi \leftarrow$ PersonalizedPageRank$(G_w, \mathbf{v})$
  else
    $\pi \leftarrow \text{LFPR}(G_w, \mathbf{v}, g_P, \varphi_{\text{walk}})$
  end if
  ▷ *6. Short-list*
  $\mathcal{C} \leftarrow$ top-L nodes by π, excluding T (and M); add the pinned set H if mode 2
  ▷ *7. Direct Monte-Carlo evaluation with common random numbers*
  draw a common set of random numbers Ξ of size R_{cand}
  $\text{sp}^0 \leftarrow$ MonteCarloSP$(G, M, T; \Xi)$
  for each $c \in \mathcal{C}$ **do**
    $\text{sp}^c \leftarrow$ MonteCarloSP$(G, M, T \cup \{c\}; \Xi)$
    $\Delta_{\text{mm}}(c) \leftarrow \min_{v \in V_{\text{vul}}} \text{sp}^c(v) - \min_{v \in V_{\text{vul}}} \text{sp}^0(v)$
    $\Delta_{\text{tail}}(c) \leftarrow \text{mean}_{v \in P} \text{sp}^c(v) - \text{mean}_{v \in P} \text{sp}^0(v)$
  end for
  $c^* \leftarrow \arg\max_{c \in \mathcal{C}} \big(\Delta_{\text{mm}}(c),\ \Delta_{\text{tail}}(c)\big)$ ▷ lexicographic: ties broken by tail gain
  $T \leftarrow T \cup \{c^*\}$
  update group-formation state (append sp to history, etc.)
end for
return T

8 Summary and Suggested Order of Implementation

Strategy	Extra inputs	Main benefit	Main risk
1. Tied-min band	$\mathrm{se}(v)$ from `sp`, R	Noise-robust, adaptive size	Band too large or too small
2. Rescuability filter	`dist`, `sp` history	Avoids stalled maximin	Drifts toward leximin
3. Community-stratified	Partition from `A`	Balanced exploration	Weak on unstructured graphs
4. MWU dual weights	Persistent λ	Principled, stable	More hyper-parameters

Suggested order. Implement **Strategy 1** first (a few lines in the current block), then **Strategy 2** as a composable filter on top of any P. Strategy 3 is worth testing on graphs with clear community structure, and **Strategy 4** is the most ambitious change and the strongest candidate for the report.

Ablation protocol. Compare, across strategies and on the same random-number streams:

- the final maximin save-probability $\min_v \mathrm{sp}(v)$,
- the lower-decile mean of sp, and
- the number of seeds with zero maximin gain.